BCS-081	Computational Geometry	L T P 3 0 0	3 Credits

Pre-requisites: students should know about the computer graphics.

Course Objectives:

CO1	Introduce rigorous algorithmic analysis for problems in Computational Geometry.
CO2	Discuss applications of Computational Geometry to graphical rendering.
CO3	Introduce the notions of Voronoi diagrams and Delaunay Triangulations.
CO4	Develop expected case analyses for linear programming problems in small dimensions.
CO5	Discuss Trapezoidal maps

Detailed Syllabus

MODULE I

Introduction, Application domains of computational geometry, Limitations of computational geometry, Convex hulls, Jarvish March method, Graham's scan method, Planar Graphs, Regions, Dual of a graph, Geometric Dual, Triangulations: polygon triangulations, guarding, Art Gallery problem.

MODULE-II

Voronoi diagrams: construction and applications, Delauney triangulations, Divide and conquer approach, Flip and incremental algorithms, duality of Voronoi diagrams, min-max angle properties. Geometric searching: point-location, Trapezoidal maps, Fractional cascading, Finger trees, Segment trees, Interval trees, Visibility: weak and strong.

MODULE-III

Arrangements of lines: zone theorem, Combinatorial geometry: Ham-sandwich cuts. Sweep techniques: plane sweep for segment intersections, Fortune's sweep for Voronoi diagrams, Topological sweep for line arrangements.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Analyze randomized algorithms for small domain problems.
CO2	Use line-point duality to develop efficient algorithms.
CO3	Apply geometric techniques to real-world problems in graphics.
CO4	Solve linear programs geometrically.
CO5	Use Jarvish March method

Text Books:

1. Franco P. Preparata and Michael Ian Shamos, "Computational Geometry: An Introduction", Springer.
2. Mark de Berg, Marc van Kreveld, Mark Overmars, and Otfried Cheong, "Computational Geometry: Algorithms and Applications", Springer.

Reference Books:

1. Ketan Mulmuley, "Computational Geometry: An Introduction Through Randomized Algorithms", Prentice-Hall.
2. Joseph O'Rourke, "Computational Geometry in C", Cambridge University Press.